Need and demand in Minnesota: where they agree, where they split, what to do on Monday

MinneMUDAC 2026 · The Food Group · graduate division. Every number is generated from the analysis outputs; no August 2026 value is used.

THREE SIGNALS, KEPT SEPARATE

Visits = `HouseholdsReg` (one trip by one household) · **Pounds distributed** = `PoundsReg` (weight leaving a site; supply as well as demand) · **Individuals served** = `IndividualsReg` (people in those households; a person recurs across visits). **Demand** is observed activity; **need** is people who lack enough food, including those who never show up, and is not measured by any dataset: we estimate it from outside data and test it.

WHAT WE FOUND

- **Q1 · Counts follow size.** Activity tracks Feeding America's food-insecure persons almost in lockstep (ρ 0.90–0.92, 2024) because both are mostly population (ρ 0.99). Per resident the link is ρ −0.02 to 0.23 (2022–24), none significant after correction. One shelf is most of a county in 32 of 86 counties, so a shelf's booked county can decide a ranking (Scott, Benton).
- **Q6 · Need score and audit queue.** Need = average percentile of food-insecurity rate, poverty rate and households without a vehicle; gap = need percentile minus activity-per-resident percentile. 14 counties are flagged, about what chance would give (16); 6 stay flagged under every test: **Chippewa, Lyon, Nobles, Swift, Todd, Wadena**, a call list, not proof they are unusual. Reported pounds reach a median 38% of their estimated meal gap (64% across counties): a restatement of the flag in meals, an upper bound.
- **Q6 · What would reject it.** The score does not rise with SNAP participation (ρ 0.66 [0.50, 0.79]: held), or not holding poverty fixed (0.28 [0.02, 0.50]: narrow pass; poverty alone is 0.62), or a household survey that includes non-users finds no more hardship where the score is high (designed, not run). Revise if ranks move under weights or inputs.
- **Q4 · Prices.** No reliable early warning: individuals served +3.0% per 1% price rise (95% 0.7 to 5.4) falls to −0.9% without February–June 2022. Visits and pounds show nothing.
- **Q5 · SNAP.** Within county agencies faster SNAP growth goes with faster visit growth (0.27 per 1%, 0.05–0.48), not clear for pounds or individuals. November 2025 (benefits not issued until Nov 7) and April 2023 (emergency allotments end) are among the biggest swings (rank 5 of 52 and 1 of 52 by size; a typical month swings ±7.5 points), but neither can be attributed and there is no larger response where SNAP use is higher.
- **Q8 · Reach.** 35% of low-income, low-access (LILA) tracts host a reporting shelf vs 24% of others. Among low-income tracts, low-access ones are also farther from a shelf: rural residents live 7.9 miles from the nearest one vs 4.1 (straight line, population-weighted); they are somewhat less likely to contain one (odds ratio 0.71, p = 0.066). 142 LILA tracts have no shelf. Counties farther from shelves record less activity per resident (ρ −0.33 visits).
- **Q7 · Forecast (appendix).** August 2026: 230,620 visits, 12,122,052 pounds; run rate of the last three months per working day weighted 1:2:3, ages added to individuals, Wilkin = 0.

MONDAY

- **Allocate.** Conversation (not review) in the six counties: hours, transport, language, unreported shelves. Blend a poverty share into the pounds split at a weight staff choose, planned by service area.
- **Site.** In LILA tracts with no shelf: extend hours or add a mobile stop before a building.
- **Argue.** Ask for advance notice of payment gaps and a de-identified SNAP–shelf linkage pilot. Shelves are not shown to absorb benefit cuts.
- **Measure.** Month-close flag, county check for large shelves (census-block county), household survey including non-users.

LIMITS

Associations, not causal effects; need is inferred; visits and individuals repeat; tribal programs and tribal-agency SNAP sit outside county rows; July 2026 completeness is unknown; the Q6 queue is for audit, not a count of unmet people.

SOURCES

Provided: Feeding America Map the Meal Gap (2024, 2025 v2, 2026 editions); BLS CPI food at home; Minnesota DCYF SNAP; Minnesota school-meal data.

Additional: Census ACS 5-year 2019–23 (B17001 poverty, B08201 vehicles, population); USDA ERS Rural-Urban Continuum Codes 2023; USDA ERS Food Access Research Atlas 2019 (LILA) and 2025 SRAM; Census 2010–2020 tract relationship, 2020 blocks and boundaries; BLS LAUS county unemployment; USDA FNS SNAP tables; 22 dated policy events (DCYF, DHS, MDE, USDA, courts); Urban Institute Well-Being and Basic Needs Survey. Citations, grain, vintage and reason chosen: Sources page and docs/sources/catalog.md (69 rows).

`Map the Meal Gap: Dewey et al. 2024; Ribar et al. 2025 (revised 28 Jul 2026), 2026. Feeding America.`